

Measurements of small-caliber ballistic shock waves in air

Roland Stoughton

Science Applications International Corporation, 10260 Campus Point Drive, San Diego, California 92121

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Megahertz bandwidth pressure measurements of airborne ballistic shock waves from bullets are presented which confirm weak-shock theoretical predictions of waveform shape, period, amplitude, and the scaling of these quantities with shock propagation distance. Wavefront distortions and amplitude variations are quantified versus shock propagation distances of 3–55 m and agree with predictions based on linear acoustic propagation through simple turbulence models. Observed rise times probably exhibit a mix of turbulence and molecular relaxation effects. © 1997 Acoustical Society of America. [S0001-4966(97)03808-3]

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INTRODUCTION

Ballistic shock waves from bullets present an interesting test of weak-shock theories and recently have been exploited, through digital signal processing, to locate snipers.^{1,2} Theoretical modeling^{3,4} of the amplitude and period of these shocks, and comparison of observations with the predicted scaling relations⁴ were done in the 1940s. More recently, Bass *et al.*⁵ have measured these shocks with an emphasis on their rise time. Rise times of these shocks,⁵ as well as those of aircraft sonic booms,⁶ are significantly larger than predicted from viscosity and still air. Atmospheric turbulence^{7–9} and molecular relaxation^{5,10,11} have been used to explain this excess, and both are established as important mechanisms. We repeat some of the earlier ballistic shock wave measurements, confirm quantitative agreement with the weak-shock predictions of shock wave period and amplitude, and also measure the effects of atmospheric turbulence on wavefront shape and peak amplitude. Good agreement is seen between our observations of wavefront spatial distortions and peak amplitude variations and the predictions of the theory of linear wave propagation through a turbulent medium.¹² This theory should be useful for predicting the conditions under which these wavefronts retain their integrity and hence when they can be exploited for practical tasks like projectile tracking and shooter localization. The linear wave theory also matches the observed rise times; however, this may be fortuitous since molecular relaxation probably is important, and since finite amplitude effects in the shock propagation may inhibit turbulent thickening of the shock leading edge.^{6,10} Our intention is not to resolve the controversy over rise times. Rather it is to present high-quality confirmatory measurements of the ballistic shock waveform dependence on miss distance, and novel measurements of the spatial coherence and amplitude variations of these shocks.

In Sec. I we summarize the predictions of weak-shock theory for a still atmosphere. Observations are presented in Sec. II and compared to these predictions. Section III discusses the effects of atmospheric turbulence and compares our observations of wavefront distortions and rise times with linear propagation theory. A discussion is given in Sec. IV, and conclusions in Sec. V.

I. WEAK-SHOCK PREDICTIONS

Several projectile diameters and farther from the trajectory, the ballistic shock wave has attained a characteristic “N” shape, as in Fig. 1. Hence the name “N wave” which is often used. This limiting form arises as fine structure in the shock wave associated with the projectile shape is swept forward in time in the case of positive pressure perturbations, and back in the case of negative pressure perturbations, into only two discontinuities. The peak pressure amplitude $p_{\max}$ and period T of the N wave are related to projectile length l , diameter d , distance from the trajectory b , and Mach number M . By modifying the linear acoustics isentropic result with the finite-amplitude flow characteristic curves, Whitham³ derives

$$p_{\max} = \frac{0.53P_0(M^2 - 1)^{1/8}}{b^{3/4}} \frac{d}{l^{1/4}}, \quad (1)$$

$$T = \frac{1.82Mb^{1/4}}{c(M^2 - 1)^{3/8}} \frac{d}{l^{1/4}}, \quad (2)$$

where P_0 is ambient pressure and c is sound speed.

Taking explicit account of energy dissipation, DuMond *et al.*⁴ derive expressions which reduce to Eqs. (1) and (2) for large b/d , but show steeper dependencies of $p_{\max}$ and T on b for smaller b/d , where “small” is not determined. Their observations do indicate a steeper dependence for the larger caliber projectiles, consistent with the form of their predictions. Atmospheric conditions, instrumentation properties, and experimental procedure were not reported by DuMond *et al.*

Rise time of the leading and trailing edges of the N wave due to viscosity is predicted to be¹³

$$t_{\text{rise}} = \frac{\lambda}{c} \frac{P_0}{p_{\max}}, \quad (3)$$

where λ is the molecular mean free path, $\lambda \doteq 8 \times 10^{-8}$ m. For typical $p_{\max}/P_0 > 10^{-3}$ in the experiments this yields rise times in the submicrosecond range. When molecular relaxation effects are included,^{5,11} the predicted t_{rise} comes out significantly larger than Eq. (3), but still in the microsecond range for small miss distances, leading to an experimental

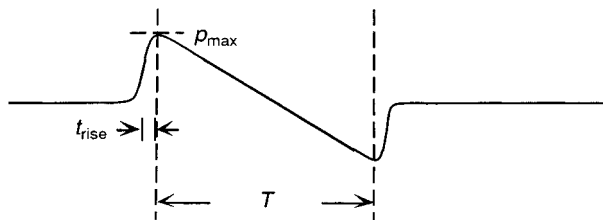


FIG. 1. Asymptotic shape of the ballistic shock wave.

requirement for piezoelectric transducers and megahertz sample rates if the shock waveform is to be resolved.

II. OBSERVATIONS OF BALLISTIC SHOCK WAVES

Data were collected in August 1995 at a local rifle range by firing past fixed condenser and piezoelectric microphones at known miss distances of 3–55 m. Data also were collected with condenser microphones at miss distances up to ~200 m at Camp Pendleton Marine Base, California, in October 1995. Although small-caliber ballistic shock waves were seen clearly out to shock propagation distances of ~300 m, quantitative analysis is limited primarily to the August data set, for which ground interaction effects were much better controlled, sound channel focusing was much less uncertain, and which was taken at a higher digital sampling rate.

In the August data set, pressure signals were burst-digitized with 12-bit precision at 2-MHz sample rate over 1-ms time windows. Experimental geometries were arranged so that all shocks arrived at near-normal incidence to the transducers' surfaces, fell within these time windows, which were triggered by a condenser microphone just ahead of the sensors to be recorded, and were free of ground interaction out to 20-m miss distance. The near-normal incidence angles were achieved through knowledge of muzzle velocity and ballistic drag coefficient, and confirmed by arrival times at our planar array. A horizontal linear arrangement of four piezoelectrics was used to measure shot-to-shot wavefront spatial distortions. Table I lists the sensor properties. The piezoelectric sensors were mounted flush with 3×3×1/4 in. aluminum baffles. Figure 2 shows raw responses for four different sensor types. The long-discharge-time piezoelectric should most faithfully represent the early part of the incident waveform. Departure of the later part of the waveform from an ideal *N* wave at this small miss distance probably is due to the finite baffle size. Intrinsic rise times of these piezoelectrics are <1 μs for normal incidence. The short-discharge piezoelectric response resembles the time de-

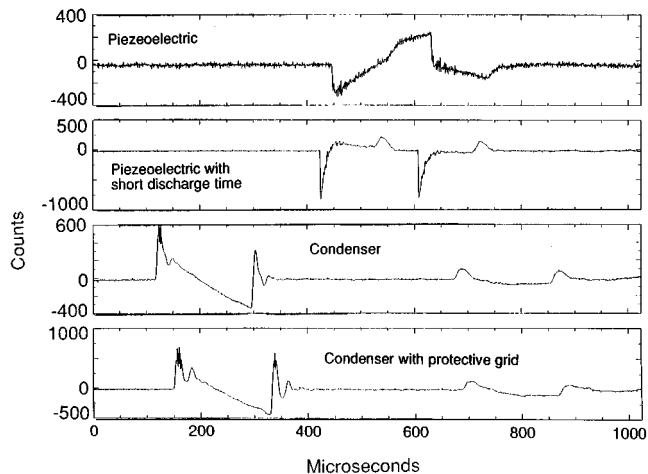


FIG. 2. Responses of different transducer types to the ballistic shock wave. 5.56-mm ammunition, 3-m miss distance. Data were digitized at 2-MHz sample rate. Transducers are described in Table I.

rivative of the long-discharge piezoelectric, and is particularly appropriate for measuring leading edge arrival times. Both have polarity reversed in Fig. 2. The two condenser microphone responses exhibit ringing associated with the detailed dynamics of the membrane, cavity, and protective grid.

Figure 3 shows piezoelectric responses for four different miss distances from 3 to 55 m, and illustrates the behavior predicted in Eqs. (1) and (2). Also obvious is the lengthening of rise time and increasing distortion of the *N* wave with increasing miss distance.

For quantitative comparison with predictions of amplitude and period, the condenser microphone responses were used since a more accurate factory calibration was available for them. Period was measured as the time between the leading and trailing edges, and amplitude as the intercept of (i) a linear fit to the linear descending portion of the waveform, and (ii) the leading edge time. These quantities were insensitive to the details of the ringing in the condenser responses illustrated in Fig. 2. Observed shock wave amplitudes and periods are summarized in Figs. 4 and 5. Behavior is very close to the nominal relations, Eqs. (1) and (2). In evaluating (1) and (2), we must calculate the Mach number at the point on the trajectory where that part of the shock wave is emitted which will eventually hit the sensors. This was done by measuring this velocity with a chronograph for several rounds from the same ammunition lot. Measured velocity varied <2%, and agreed with Remington's ballistic data. Emission Mach was 2.7 for the 5.56-mm-diam and 1.9 for the 7.62-mm-diam projectiles, for the ambient sound speed of

TABLE I. Properties of the different transducers.

Sensor	ACO 7017	PCB 102A07	PCB 132A40
Type	1/4 in. condenser	quartz piezoelectric	ceramic piezoelectric
Dynamic range	50–164 dB	131–205 dB	115–171 dB
Bandwidth	5–70 000 Hz	1–500 000 Hz	10 kHz–1 MHz (Long discharge time) 60 kHz–1 MHz (Short discharge time)
Rise time			
Normal	~2 μs	2 μs	<1 μs
Oblique	25 μs	8 μs	3 μs

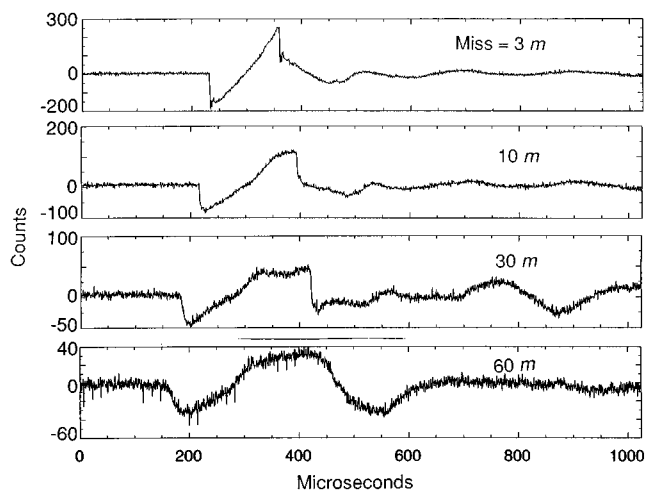


FIG. 3. Evolution of ballistic shock wave with miss distance. 5.56-mm ammunition, miss distances 3–55 m. Vertical scale (ADC counts) changes from plot frame to plot frame.

355 m/s (40 °C). In addition to a small bias, the amplitude behavior for the 5.56-mm projectiles shows a significant departure from the $b^{-3/4}$ power law at the larger miss distances. This could be due to sound channel focusing. The propagation path was 2 m off the ground on a hot day with solar heating. The sound-speed profile $c(z)$ was observed to be upward refracting and of curvature $\partial^2 c / \partial z^2$ in the sense that

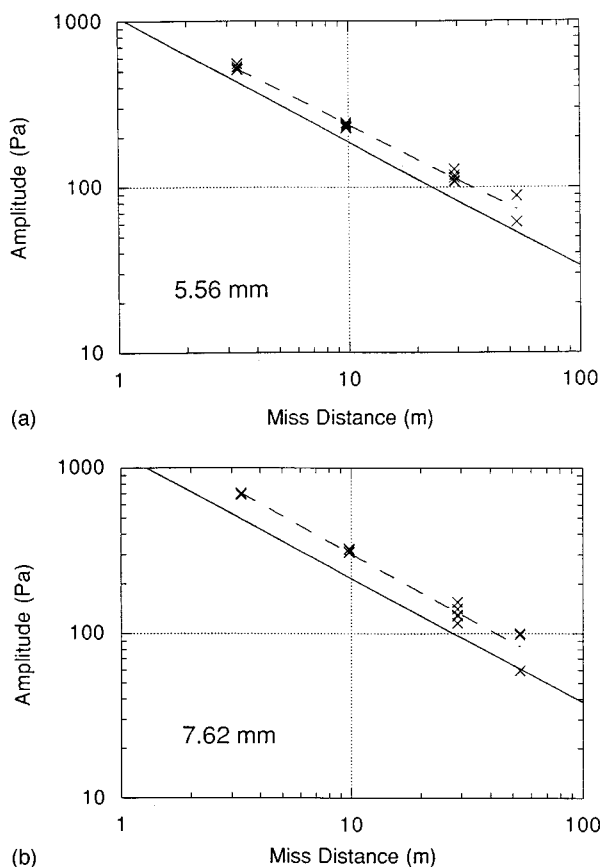


FIG. 4. Peak amplitude of N wave versus miss distance. (a) 5.56-mm ammunition, (b) 7.62-mm ammunition. Dashed lines are best fits to data. Solid lines are predictions from Eq. (1). Numbers of data points varied from 3 to 6 at any miss distance.

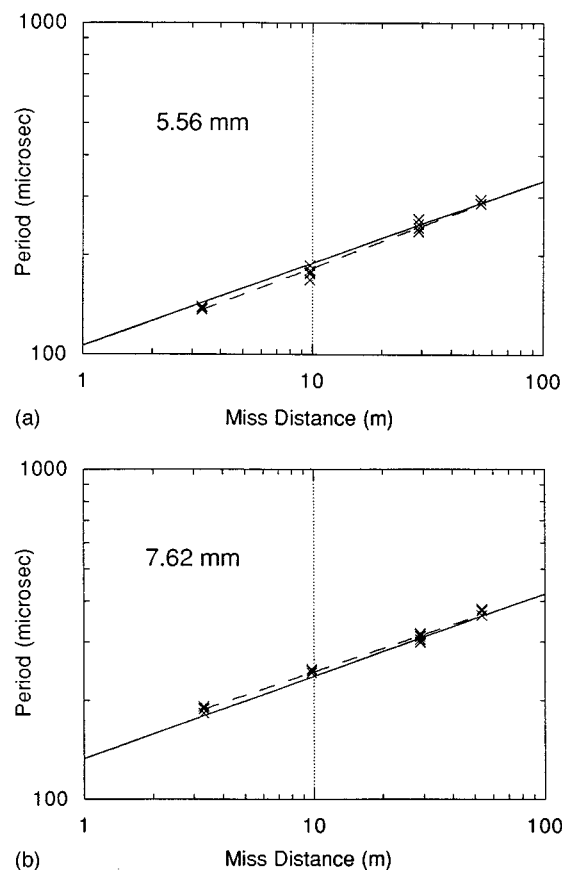


FIG. 5. Period of N wave versus miss distance. Dashed lines are best fits to data. Solid lines are predictions from Eq. (2).

would cause excess received amplitude, with this excess increasing with miss distance. The influence of $c(z)$ was not addressed by DuMond *et al.* Without accurate synoptic measurements of wind shear and temperature along the propagation path, we can only produce an order-of-magnitude estimate of the effect. Taking the observed temperature values of $\partial c / \partial z \doteq -0.3 \text{ s}^{-1}$, $\partial^2 c / \partial z^2 \doteq 0.3 \text{ m}^{-1} \text{ s}^{-1}$ gives an intensity focusing which reaches double the free-field value at $R = 100 \text{ m}$, and the excess intensity scales as the square of propagation distance. Also, ground interaction becomes significant at the 30- and 55-m miss distances. We therefore distrust the data points at $R = 30 \text{ m}$ and $R = 55 \text{ m}$ when trying to make an accurate determination of the amplitude scaling with range. Using the points at $R = 3$ and 10 m , however, should keep the focusing effects at or below the 1% level, and should have no ground reflected paths in the data time window.

The effects of propagation uncertainties on period T are more complex. Again, we may take the 3- and 10-m points as most reliable. The resulting estimates for the power-law exponents are given in Table II. Determinations are consistent with the asymptotic rules $P_{\max} \sim b^{-3/4}$ and $T \sim b^{1/4}$ found by Whitham³ and DuMond *et al.*⁴

Figure 4 shows agreement not only with the predicted scaling relations, as was shown by DuMond *et al.*,⁴ but with the constant coefficients in Eq. (1), to within the $\pm 30\%$ experimental uncertainties in amplitude calibration. Figure 5 shows small systematic departures of measured periods from

TABLE II. Estimates of power-law exponents for $p_{\max} \sim b^n$ and $T \sim b^m$. Data at 3- and 10-m miss distance only were used. Uncertainties represent 90% confidence intervals, based on scatter of repeated measurements at each miss distance.

Ammunition	5.56 mm	7.62 mm
n	-0.738 ± 0.027	-0.734 ± 0.015
m	0.230 ± 0.018	0.249 ± 0.012

Eq. (2) at 3- and 10-m miss distances. But these are in the opposite sense for the two projectile sizes. We have been unable to explain them as a measurement bias.

To compare measured rise times with Eq. (3) we use the long-discharge-time piezoelectric responses and define observed rise time to be the ratio of maximum pressure to maximum rate of increase of pressure, where this rate was estimated by least-squares line-fitting the portion of the leading edge between 10% and 90% of peak amplitude. This procedure was found to be unbiased with respect to, and gave smaller variance in repeated measurements than, the 10%–90% rule with smoothing. Figure 6 shows that the observed rise times are an order of magnitude longer than the weak-shock prediction, Eq. (3). As mentioned in Sec. I, a similar discrepancy was noted in the 1970s for aircraft sonic booms;^{6,7} molecular relaxation and propagation through atmospheric turbulence both contribute in amounts that depend on the exact geometry and environmental conditions of the propagation.^{6–9,14} Reference 5 attempts to match small-caliber ballistic shock rise times with molecular relaxation, obtaining order-of-magnitude agreement. However, their agreement is the worst for their indoor data set where turbulence is effectively absent. Deducing the physical mechanism is complicated by the roughly similar trend of rise time with propagation distance (decreasing peak pressure) predicted from molecular relaxation to that predicted from the accumulation over distance of turbulence effects. Further discussion is given in Sec. IV.

III. EFFECTS OF ATMOSPHERIC TURBULENCE

Our measurements support estimation of wavefront distortions down to the $\sim 1\text{-}\mu\text{s}$ level, and amplitude variations

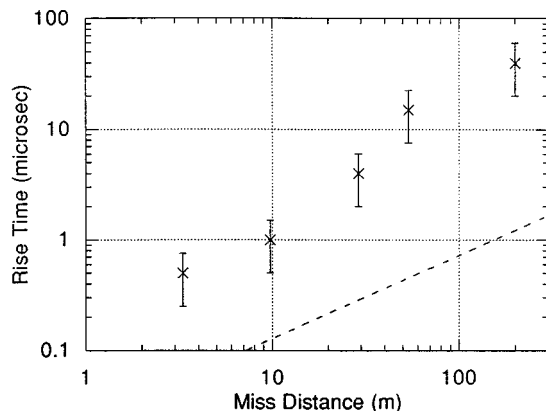


FIG. 6. Rise time of N wave versus miss distance, 5.56-mm ammunition. Dashed line is prediction from Eq. (3), which assumes a homogeneous atmosphere. The observation at 200-m miss distance comes from the October 1995 data set with condenser microphones and 100-kHz sampling.

to the $\sim 10\%$ level. Here we present these estimates and interpret them in terms of the effects of atmospheric temperature and velocity fluctuations, using a propagation theory appropriate to linear waves and stationary atmospheric fluctuation statistics. In our experiment, the short horizontal propagation path suggests a stationary model. Discussion of nonlinear propagation effects is deferred to Sec. IV. We also compare predictions of this theory to the observed rise times, although this comparison is complicated by both nonlinearity and molecular relaxation.

Levels of turbulence were not measured during the data collection, but we can make predictions with an assumed spectrum using its level as a free parameter. We find that the observed wavefront distortions, amplitude variations, and lengthened rise times, and their dependence on shock propagation distance all are predicted by the same turbulence strength and spectral shape.

A. Turbulence model

We adopt a turbulence energy spectrum over isotropic wave number

$$F(k) \sim k^{-p}, \quad \lambda_0 < k^{-1} < L, \quad (4)$$

where λ_0 , L are the inner and outer scales of the turbulence. L also is identified with the correlation length. Typical observed correlation behavior and fractional mean-squared sound-speed fluctuations μ^2 near the ground indicate^{15,16}

$$L \sim 1 \text{ m}, \quad 10^{-7} < \mu^2 < 10^{-5}, \quad 5/3 < p < 2,$$

where $p=5/3$ is the Kolmogorov spectrum and $p=2$ arises when the turbulence is composed of random vortex sheets. For this range in the exponent p , results are not very sensitive to the inner scale λ_0 .

B. Saturated and unsaturated regimes

The mechanics of converting a turbulence model to predictions of wavefront distortions are well established.¹⁷ A very condensed discussion will be repeated here. The controlling parameters, usually denoted Φ and Λ are, respectively, the rms phase fluctuation due to sound-speed fluctuations integrated along a line from source to receiver, and a measure of the average number of independent fluctuations present in a Fresnel zone centered on that line. We have¹⁷

$$\Phi^2 = q_0^2 \mu^2 R \int_{-\infty}^{\infty} \rho(u) du, \quad (5)$$

$$\Lambda = R/6L^2 q_0, \quad (6)$$

where $q_0 = 2\pi f/c$, f is frequency, R is the propagation distance, and $\rho(u)$ is the correlation function, normalized so that $\rho(0) = 1$. For $p=5/3$, Eq. (5) becomes

$$\Phi^2 \doteq 0.4 q_0^2 \mu^2 RL. \quad (7)$$

The most fundamental distinction in analyzing the propagation for any R , f , p is between “unsaturated” regimes, where $\Phi \Lambda^{p/4} < 1$ and geometrical optics is a good approximation, and “saturated” regimes, where $\Phi \Lambda^{p/4} > 1$ and significant amplitude fluctuations and waveform distortions

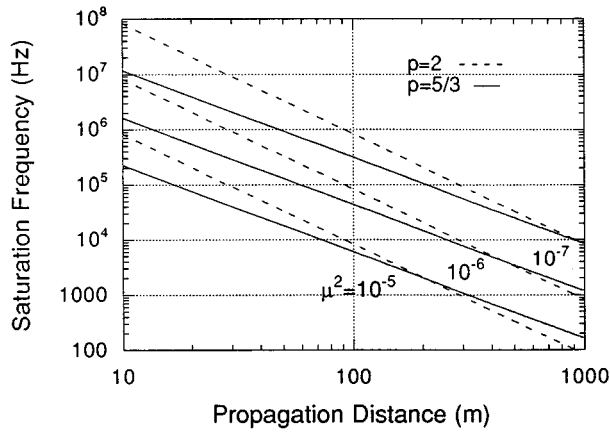


FIG. 7. Boundaries between saturated and unsaturated propagation, from Eqs. (8) and (9). Saturation occurs above the line. Solid lines are for turbulence spectral power-law index $p=5/3$, dashed lines are for $p=2$. $c=340$ m/s was used.

occur.¹² This distinction can be expressed as a saturation frequency, f_{sat} using (6) and (7),

$$p=5/3 \rightarrow \Phi^2 \Lambda^{5/6} = 1 \rightarrow f_{\text{sat}} \doteq 1.3cL^{4/7} \mu^{-12/7} R^{-11/7}, \quad (8)$$

$$p=2 \rightarrow \Phi^2 \Lambda = 1 \rightarrow f_{\text{sat}} \doteq 2.5cL \mu^{-2} R^{-2}. \quad (9)$$

This is illustrated in Fig. 7. Using $\mu^2 = 10^{-6}$ for example we see that at propagation distances >60 m the $\sim$ MHz frequencies associated with the shock discontinuities will be saturated. The slight distortions of the entire waveform seen in Fig. 2 at the larger miss distances also are consistent with an approach to saturation for frequencies of ~ 10 kHz at ranges ~ 60 m indicated in Fig. 7.

C. Wavefront coherence

Because the dominant wavelengths in the shock waveform are clearly unsaturated at $R < 60$ m, we may use geometrical optics approximations to compute arrival time variance and covariance between sensors. For separations transverse to the propagation direction we find arrival time covariance¹⁷

$$D_t^2 \equiv E[(\tau(x) - \tau(x + \Delta x))^2] = 2.2\mu^2 \frac{RL}{c^2} \left(\frac{\Delta x}{L} \right)^{5/3}, \quad (10)$$

$$\lambda_0 \ll \Delta x \ll L$$

$$= 0.8\mu^2 \frac{RL}{c^2}, \quad \Delta x \sim L, \quad (11)$$

where these have been derived for the case $p=5/3$.

These two precise limiting forms can be joined with the approximation

$$D_t^2 = 2\sigma_t^2(1 - \rho_t(\Delta x)), \quad (12)$$

where

$$\sigma_t^2 = 0.4\mu^2 \frac{RL}{c^2}, \quad (13)$$

$$\rho_t(\Delta x) = (1 + 2.75(\Delta x/L)^{5/3})^{-1}. \quad (14)$$

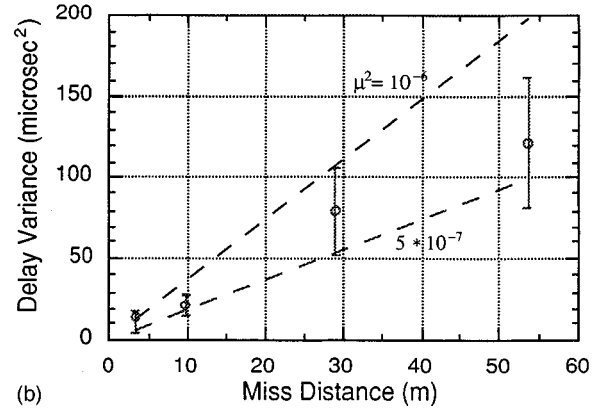
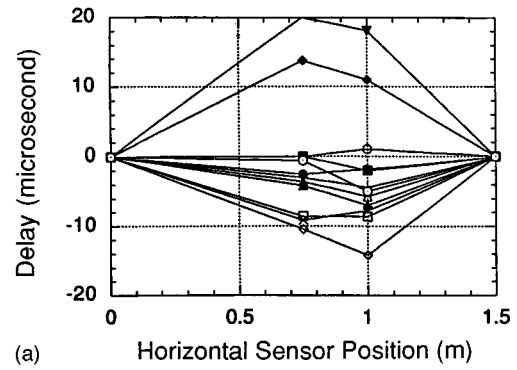


FIG. 8. (a) Residual shock wavefront distortions with respect to a plane wavefront passing through the two end-point sensors in a four-sensor horizontal array. 30-m miss distance. (b) Combined variance of the two middle sensors' residual arrival times versus miss distance. Dashed lines are predictions from Eq. (12), for two different assumed turbulence strengths.

σ_t^2 represents the total variance in acoustic travel time along a ray path from source to receiver; $\rho_t(\Delta x)$ is the correlation of this delay in the transverse direction.

Equations (10) and (11) were derived for a point source and receiver. Our shock wave is generated by a moving source and exists as an expanding conical wavefront with the projectile at the vertex. Coherences over separations in a plane containing the trajectory therefore may be better predicted based on a line source than a point source. This would affect the numerical coefficients in Eqs. (10), (11), (13), and (14). However, conclusions will be insensitive to this change since we will infer μ^2 from the data. Therefore the calculations in Ref. 17 have not been redone for a line source or moving point source.

Observations of horizontal coherence were done by firing repeated shots at the same nominal geometry about 2 s apart past a 1.5-m horizontal array of four short-decay piezoelectric sensors. The 2-s interval was chosen as somewhat longer than the turbulence decorrelation time for scales ~ 1 m in the light winds (1 to 2 m/s) present during the data collection. To compare observations with theory, the shot-to-shot variation in wavefront tilt had to be removed from both, since in the data this was contaminated with shot-to-shot variation in bullet velocity and trajectory. This was done by replacing the four arrival times with their residuals from a line through the two end-point sensor arrival times. The two end-point times then become zero [Fig. 8(a)]. The variances of the two nonzero times were then compared to predictions

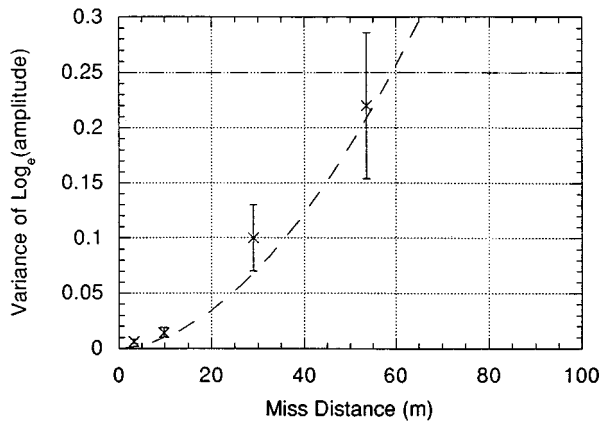


FIG. 9. Amplitude variations compared to predictions based on the turbulence spectral level $\mu^2 = 7 \times 10^{-7}$ inferred from Fig. 8. The saturation boundary is at a vertical coordinate of 0.25.

[Fig. 8(b)], where the predictions used Eqs. (12)–(14) to express these variances as a linear transformation of the covariance between the four sensors. The inferred level of μ^2 is 7×10^{-7} , consistent with other observations in similar conditions.¹⁵ Our data were collected over sand in full sun during late-morning hours, in very light winds, with an ambient temperature of $\sim 40^\circ\text{C}$.

D. Amplitude fluctuations

Using the spectral level inferred in Sec. III C from the arrival time variations we can predict the amplitude fluctuations and rise times. Variance of log (amplitude) is related to Λ , Φ as¹⁷

$$\sigma_{\log a}^2 = \frac{1}{4} \Lambda^{p/2} \Phi^2 \quad (15)$$

and is equal to 1/4 at the boundary between saturated and unsaturated regimes. Using (5) and (6) we find

$$\sigma_{\log a}^2 = 0.022 L^{-2/3} \mu^2 R^{11/6} (2\pi f_0 / c)^{7/6}, \quad p = \frac{5}{3}, \quad (16)$$

which depends explicitly on frequency. By using data from the short-discharge-time piezoelectrics, we can arrange a well-defined characteristic frequency f_0 for the observed signals so that data may be compared with (16). The frequency response of these piezoelectrics is bounded below by the discharge time and above by the rise time. Defining f_0 as the centroid of the frequency response we are able to derive

$$f_0 = t_d^{-1} \ln(t_d / t_r), \quad (17)$$

where t_d is the (exponential) discharge time and t_r the sensor response rise time. Examination of typical sensor responses to shocks at small miss distance gives $t_d = 4 \mu\text{s}$, $t_r = 0.5 \mu\text{s}$. Substituting Eq. (17) into Eq. (16) we compare with data as shown in Fig. 9, where amplitude has been measured as the peak in the response, and the curve represents Eq. (16). Evidently the predictions based on spectral level inferred from the wavefront coherence measurements are a good fit to the observed amplitude fluctuations.

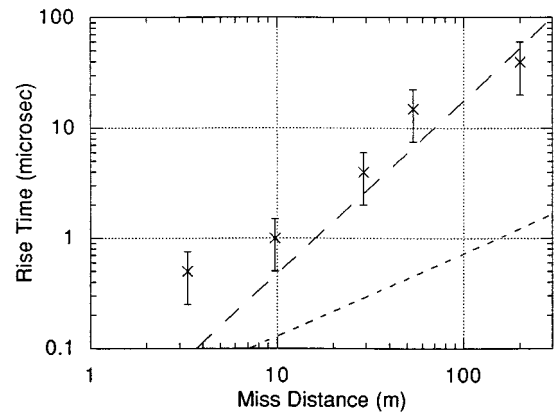


FIG. 10. Rise time versus predictions based on still air (short dash) and turbulent air (long dash) with turbulence level inferred from Fig. 8.

E. Rise times

A rough prediction of shock waveform rise times due to turbulence can be made by arguing that all frequencies greater than the saturation frequency have their relative phases scrambled and cannot contribute to a sharp leading edge. This suggests

$$t_{\text{rise}} \doteq f_{\text{sat}}^{-1} \quad (18)$$

with f_{sat} from Eq. (8). More sophisticated calculations of linear-wave pulse spreading are possible,¹⁸ but are not supported by our environmental data. The predictions using Eqs. (8) and (18) are plotted with the rise-time data in Fig. 10. Here again we set $\mu^2 = 7 \times 10^{-7}$ as inferred from the wavefront coherence measurements, and take $L = 1 \text{ m}$. At the shortest miss distance the observed rise time is set by the sensor response rise time of $\sim 1/2 \mu\text{s}$. The data point at the largest miss distance is from an October 1995 data set, in which propagation was across a valley, farther above the ground and therefore in weaker turbulence than the other data points which were from the August data set. The prediction therefore fits the data quite well where it should, and much better than the predictions for still air, Eq. (3).

IV. DISCUSSION

The consistency of the observed wavefront distortions and amplitude variations with turbulence predictions using only the one free parameter μ^2 suggests that the theory is appropriate for these observables and that the actual sound-speed variations had $\mu^2 \sim 7 \times 10^{-7}$. Changing the assumed spectral index p , and correlation length L within reasonable limits would have changed the inferred μ^2 , but would not have affected the agreements between theory and data significantly.

The successful prediction of rise times may be fortuitous, since molecular vibrational relaxation effects are predicted to be significant, and since the wave is nonlinearly bound. The only data set comparable to ours which eliminates the effects of turbulence is that from “range 4” given in Ref. 5, which was an indoor range. Those rise-time data do not correlate well with the authors’ predictions based on vibrational relaxation. The authors attribute this to an instru-

mental rise-time response limit of $2\ \mu\text{s}$. However, a limit of $2\ \mu\text{s}$ does not explain the discrepancy between their “range 4” data and the theory. More recent modeling of rise time¹¹ coupling the nonlinearity with molecular relaxation shows longer rise times and large sensitivity to humidity (fraction of air made up of water molecules), which might be capable of explaining our observations without turbulent spreading.

As pointed out in Ref. 5, nonlinearity of the shock wave leads to resharping on length scales which should be much shorter than the distances over which we are hypothesizing accumulation of turbulence effects. This argument would seem to invalidate our linear-wave prediction of rise times in turbulence. It does not undermine the turbulence explanation for the wavefront distortions and amplitude variations, since even finite-amplitude wavefronts are advanced, retarded, and focused by variations in local ambient sound speed. It is beyond the scope of this paper to try to resolve authoritatively the issue of the relative importance of turbulence and molecular relaxation in the rise times. In the context of aircraft sonic boom rise times, nonlinear⁷ and nonstationary^{6,9} propagation models have been employed, with order-of-magnitude success, and numerical prediction codes are available that show good agreement with each other and with at least some data sets.¹⁴

V. CONCLUSIONS

Measurements of small-caliber ballistic shock wave amplitude and period were made that confirm predictions based on weak-shock theory. Both the power-law exponents (Table II) and the constant coefficients agree closely with predictions. Despite the nonlinear nature of the shocks, the theory of linear acoustic wave propagation in turbulence is able to model successfully, with only one free parameter, the observed wavefront distortions, amplitude variations, and rise times of these shock waves, although the rise times would be more defensibly modeled with a combination of nonlinear wave propagation and molecular relaxational effects.

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